

Adapted Coherent Weighting in Photoacoustic Tomography

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Abstract— In this paper, we proposed an adapted coherent factor (ACF), which is applied to a circular scanning photoacoustic computed tomography (C-PACT) arrangement. For validation, ACF is applied to different reconstruction algorithms such as universal-back projection (UBP) and adapted delay and sum (ADAS) with two types of SNR levels (80 dB and 20 dB) and 2 types of detectors with different bandwidths (infinite bandwidth and 2.25 MHz central frequency, 70% bandwidth). Simulation results indicate ACF can improve the Full width at half maximum (FWHM) of the imaging system point-spread function (PSF) and contrast ratio (CR) by 16% - 20% and 13.68-30.24 dB, respectively. These quantitative values are different due to the different reconstruction algorithms, SNR levels and detector bandwidth.

Keywords— Adapted coherent factor, Circular scanning photoacoustic tomography, Contrast enhancement, Point spread function, Point-like detectors.

I. INTRODUCTION

In the last two decades, especially in recent years, photoacoustic imaging (PAI) has made great strides. The basis of photoacoustic was first given by Alexander Graham Bell as the photoacoustic effect in 1880 [1]. PAI is a hybrid imaging modality that works with a combination of light radiation and sound detection. Since body tissues have a high optical absorption coefficient, they are able to produce a strong photoacoustic signal that will cause a high-quality image, which means that the combination of light and sound will result in a good quality imaging performance compared to other imaging modalities [2-4].

Photoacoustic imaging can be divided into two categories: photoacoustic microscopy (PAM) and photoacoustic tomography (PAT). In photoacoustic tomography if the detection system has an array of transducers, it is called photoacoustic computed tomography (PACT). One of the most widely used geometries for PACT, is cylindrical/circular PACT, whose equations are described in [5, 6]. Many methods for image reconstruction in PAI have been proposed so far that have tried to improve the reconstructed image quality in both PAT and PAM in view of signal-to-noise ratio (SNR), image contrast and resolution. In PAT, modified delay and sum beamformer (MDAS) was proposed for improving tangential resolution [7, 8]. Some other studies use a positive/negative focused transducer or an acoustic lens for improving views or acceptance angle of detection geometry to improve reconstructed image [9, 10]. Multi angle photoacoustic tomography (MA-PAT) system

was proposed to compensate for the transducer finite-size in PAT and to increase the reconstructed image quality [11, 12]. Moreover, some deep-learning and iterative approaches are proposed for improving SNR and resolution in PAT system [13, 14]. In other researches, virtual detector concept was proposed to enhance contrast and tangential resolution in PAT and PAM structures [15-17]. Most of the mentioned methods are designed for finite element transducers and are mathematically complex and time consuming. In addition, none of them is used for point-like detectors.

Coherent factor weighing as a method of improving SNR is one of the fastest and most effective weighing factors, most widely used in PAM [18], which calculates the coherent to incoherent ratio in each A-line to be multiplied by the main signal as a weighing factor. This method has also been reported in some photoacoustic tomography papers with linear array detection geometry, which has been used to increase the quality of the reconstructed image in a planar geometry [19].

In this paper, we apply CF to circular scanning photoacoustic computed tomography (PACT) with point detectors. Due to the use of point detectors, there are overlaps of different A-lines in each location, so the coherent to incoherent ratio at each stage of back-projection can be calculated, which is called adapted coherent factor (ACF). The advantage of this method is that it can be used for all geometries and boosts image SNR and contrast ratio (CR), whether in limited-view or full-view circular scanning geometries and any other desired structures. To validate this method, ACF is applied on several samples of different 2D geometries that are simulated and reconstructed by two different beamformers such as adapted delay and sum (ADAS) [20] and Universal back-projection (UBP) methods [21].

In section 2 the theory of reconstruction algorithm including ACF weighting is presented. The simulation study and results are given in section 3. While the discussion and conclusion are presented in section 4.

II. MATERIALS AND METHODS

In photoacoustic phenomena, the acoustic pressure at $p(r, \bar{t})$ is determined by optical absorption properties $\phi(r')$ and shape of the irradiating pulse $\eta(r')$ [22]:

$$p(r, \bar{t}) = \frac{I_0 \beta \vartheta_s}{4\pi C} \iiint \varphi(r') \frac{d\eta(r')}{dt'} \frac{dr'}{|r-r'|}. \quad (1)$$

Where β is volume thermal expansion coefficient, ϑ_s is speed of sound and C is the specific heat. Moreover r' and r are the distances from absorber and acoustic pressure detector from the scanning center, respectively.

In reconstruction procedure, each detector senses acoustic pressure $p_n(\bar{t})$, where n represents the number corresponding to the location of the detector. A schematic for ADAS/UBP beamformer with initial Derenzo phantom (see Fig. 1(a)) is indicated in Fig. 1(b). In these algorithms for point like detectors it can be assumed that each detector can receive photoacoustic beam in any direction. In ADAS/UBP algorithm, given that the direction of the received beam cannot be distinguished by the detectors, each processed signal is back-projected in all directions. By summing all back-projected signals (see image slices in Fig. 1(b)), the final reconstructed image will be obtained.

It can be concluded that the intensity of each pixel $p_0^{(b)}(x, y)$ in the ROI is the sum of the related processed photoacoustic signals received by detectors from that particular pixel which is given in (2) for ADAS and (3) for UBPP beamformer.

$$p_0^{(b)}(x, y) = \sum_{i=1}^N p_{n, ADAS}(x, y) \quad (2)$$

$$p_0^{(b)}(x, y) = \sum_{i=1}^N p_{n, UBPP}(x, y) / \sum_{i=1}^N \Delta\Omega_i \quad (3)$$

Where N is the number of detection points, $\Delta\Omega_i$ is a weighting factor based on the solid angle for a detection element dS_0 with respect to a reconstruction point, $p_{n, ADAS/UBPP}(x, y)$ is the back-projected signal of initial pressure distribution from the n th detector, which are given in (4) and (5) for ADAS and UBPP, respectively. Moreover, the values of x and y in (2) and (3) are the coordinates of the initial pressure reconstructed at any point in the imaging ROI.

$$p_0^{(b)}(x, y) = 2p_n(\bar{t}) - 2\bar{t} \frac{\partial}{\partial \bar{t}} p_n(\bar{t}) \quad (4)$$

$$p_0^{(b)}(x, y) = p_n(\bar{t}) / \bar{t} \quad (5)$$

In point-like detector scenario, each pixel in ROI, is in field of view (FOV) of all detectors. Based on this assumption, the defined ACF for the pixel in (x, y) location is the sum of the squares of this point back projected in each direction over the square sum of all back-projected signals in different directions which is the coherency to the incoherency ratio of each pixel in each back-projection procedure (6).

$$ACF(x, y) = \frac{\sum_{n=1}^N |p_n(x, y)|^2}{N |\sum_{n=1}^N p_n(x, y)|^2} \quad (6)$$

Multiplying (6) by the initial pressure reconstructed in (2, 3) will result in (7).

$$p_{TCF}(x, y) = p_0^{(b)}(x, y) * ACF(x, y) \quad (7)$$

This is the final reconstructed image which is expected to be of higher quality compared to the conventional reconstruction result. This superior performance will be validated through simulations in the following section.

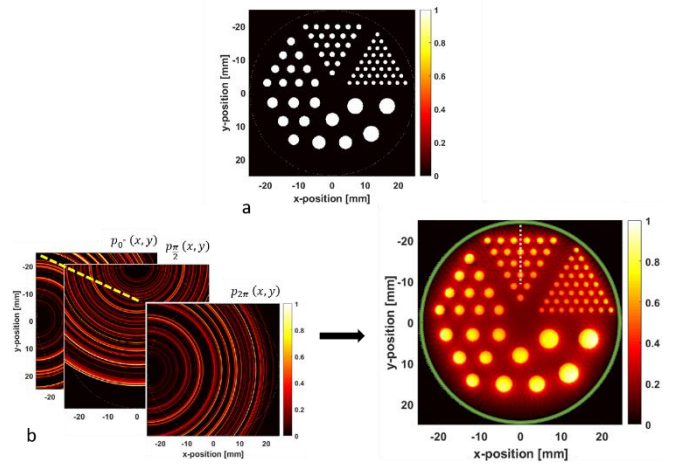


Fig. 1: Schematic diagram of ADAS/UBP beamformer. (a) The initial numerical phantom. (b) The beamformer algorithm by summing all back-projected signals (image slices), final reconstructed image.

III. SIMULATION STUDY AND RESULTS

A. Simulation Study

As mentioned in the previous section, the here-defined ACF can be used for any desired transducer geometry. Since CF was used extensively in linear structures and was not used in other structures, the linear structure was not simulated and 2D circular geometry was simulated. In the following simulations, k-wave toolbox [5] is used and a homogeneous medium with a sound speed of 1500 m/s is considered. The medium size was 5 cm at both the tangential and radial (520×520 grids) directions and sampling frequency was set at 100 MS/s. 360 point-like detectors are arranged around the ROI with a scanning radius of 2.5 cm (see green circle in all Figs. 2-4). For reconstruction, both ADAS and UBPP are used. In subsection B, it is assumed that the bandwidth of detectors is infinite and Gaussian noise with SNR level of 80 dB is added to all simulations. In subsection C, 360 infinite bandwidth detectors are used. Moreover, Gaussian noise with SNR level of 80 dB is added to all simulations. In subsection D, 360 detectors with central frequency of 2.25 MHz and 70% bandwidth were used. Again, Gaussian noise with SNR level of 20 dB is added to all simulations. For a better comparison of the reconstructed images, we set the negative parts of the reconstructed image in Fig. 2-3-4 (a, c, e, g) to zero and For Figs. 2-4 (b, d, f, h), first Hilbert is taken, then the absolute value is taken, and finally the result is normalized and indicated in dB level. But in Fig. 2-4 (i), all the positive and negative parts of the indicated A-line are plotted. Derenzo phantom is used for all simulations in this paper. For quantitative comparison, contrast ratio (CR) which is given in (8) is reported for one-dimensional (1-D) diagram of indicated dashed line (see yellow dashed-line in Fig. 2-4(a, c, e, g)).

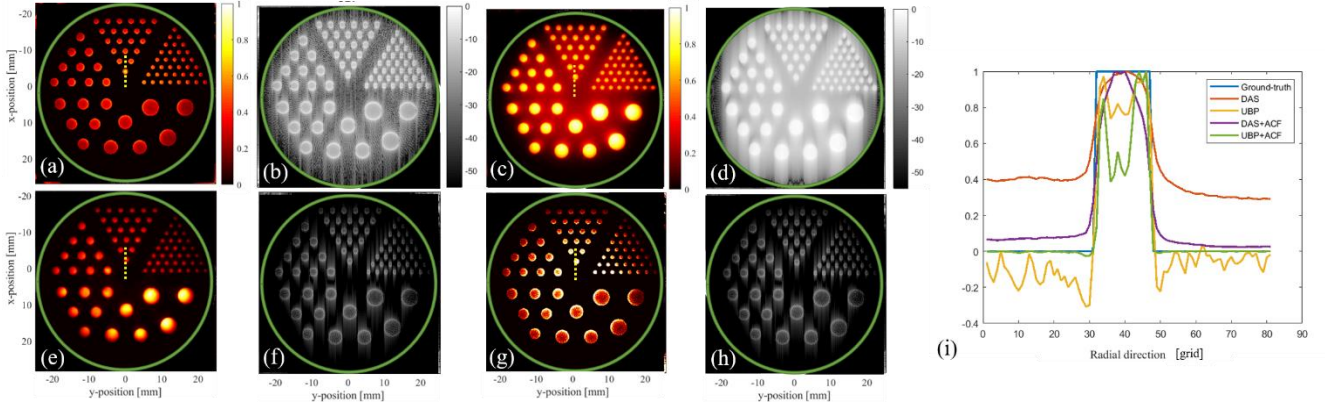


Fig. 2. Simulation results of Infinite bandwidth detector and 80 dB SNR level scenario. The initial Derenzo phantom and 360 point-like detectors with full-view circular arrangement. (a, c) The reconstructed image with UBP and ADAS methods, respectively and are shown in (b, d) in dynamic range of 55 dB. (d, e) reconstructed image with applying ACF to UBP and ADAS, respectively and also, are shown in (f, h) in dynamic range of 55 dB. (i) 1-D plot of PA amplitude of dashed-lines indicated in (a).

$$CR = 20 \log\left(\frac{\max(\text{signal})}{\text{mean}(\text{back-ground})}\right) \quad (8)$$

B. Infinite bandwidth detector and 80 dB SNR level scenario

In infinite bandwidth and high SNR level scenario, reconstruction with UBP has better contrast ratio than the ADAS. This is because the process in the UBP algorithm can be considered as a ramp filter in the frequency domain, which degrades the DC and low frequency components, and removing these components will cause better contrast in the image (see Fig. 2(a, c) and Fig. 2(b, d) in dB mode). Applying ACF to UBP and ADAS will improve contrast ratio (see Fig. 2(e, g) and also, compare (b, d) with (f, h)). For better comparison and evaluation of the proposed ACF method, 1-D diagram of the indicated dashed line (Fig. 2(a, c, e, g)) is considered (Fig. 2(i)). Based on the 1-D-line ADAS beamformer it can be seen that ADAS beamformer will increase back-ground artifacts, which is much less in UBP, but UBP has negative artifacts. Adding the proposed method to ADAS and UBP algorithm will reduce artifacts (both positive of the ADAS method and negative of the UBP method) and will also increase the contrast of the image. The quantitative performance values of indicated dashed-line are given in Table 1. Moreover, The FWHM of the imaging system's point-spread-function (PSF) without applying ACF to ADAS/UBP is 0.164 mm for both ADAS and UBP algorithms and with applying ACF to ADAS/UBP is about 0.124 mm for both of them for a point source located near the center. It can be concluded from this subsection that applying ACF improves the FWHM of the imaging system's PSF by more than 20% and reduces the resultant artifacts in the reconstructed image. The SNR value in reconstruction with ACF is 30.24 dB better than the reconstruction results without ACF, for UBP method. Also, for ADAS+ACF, it is 15.46 dB better than the results of reconstruction without ACF.

TABLE 1: Quantitative values (CR) for infinite bandwidth and 80 dB SNR level scenario.

Beamformer	UBP	ADAS	UBP+ACF	ADAS+ACF
CR (dB)	19.86	8.54	50.1	24

C. Infinite bandwidth detector and 20 dB SNR level scenario.

In the infinite bandwidth and low SNR scenario, like the previous subsection, reconstruction with UBP has a better contrast ratio than ADAS (see Fig. 3(a, c) and Fig. 3(b, d) in dB level). Applying ACF to UBP and ADAS will improve the contrast ratio of both methods (see Fig. 3(e, g) and also, compare (b, d) with (f, h)). For better comparison and evaluation of the ACF method, a 1-D diagram of indicated dashed line (Fig. 3(a, c, e, g)) is considered in Fig. 3(i). Based on the demonstrated 1D-lines, it can be seen that the ADAS method will increase background artifacts, which is much less in UBP, but UBP has negative artifacts. Adding the proposed method to the ADAS and UBP algorithm will reduce artifacts (both positive artifacts of the ADAS method and negative artifacts of the UBP method) and also increase the image contrast. The quantitative values of indicated dashed line are given in Table 2. Moreover, The FWHM of the imaging system's PSF without applying ACF to ADAS/UBP is 0.14 mm for both ADAS and UBP algorithms, and applying ACF to ADAS/UBP is about 0.12 mm for both of them for a point source located near the center. It can be concluded from this subsection that applying ACF improves the FWHM of the imaging system's PSF by more than 16% and reduces the resultant artifacts in the reconstructed image. The SNR value in reconstruction with ACF is 20.5 dB better than the results of reconstruction without ACF for UBP and ADAS+ACF, this improvement is 15.63 dB.

TABLE 2: Quantitative values (CR) for infinite bandwidth and 20 dB SNR level scenario.

Beamformer	UBP	ADAS	UBP+ACF	ADAS+ACF
CR (dB)	16.23	8.53	36.73	24.16

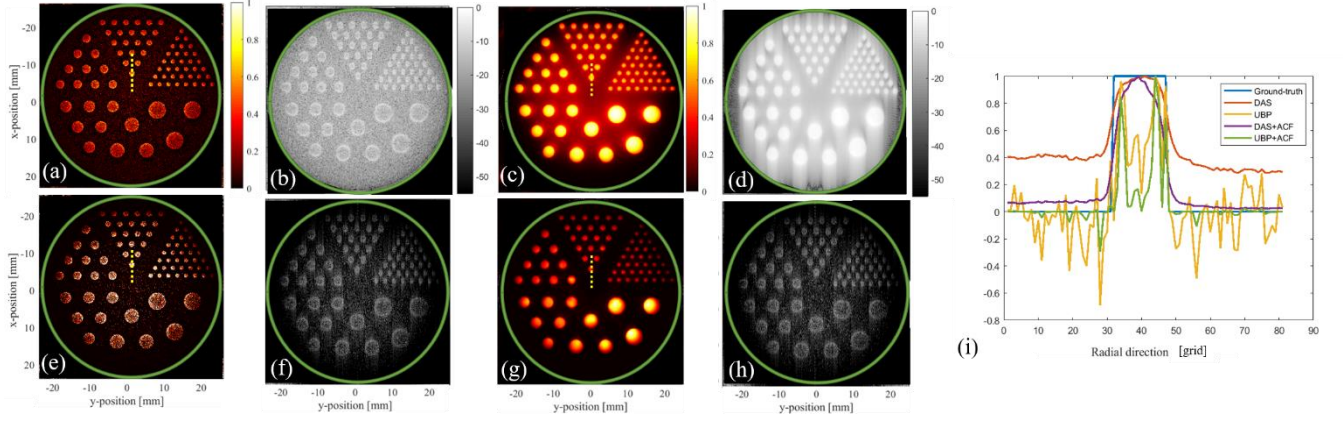


Fig. 3. Simulation results of infinite bandwidth detector with 20 dB SNR level scenario. (a) initial Derenzo phantom and 360 point-like detectors with full-view circular arrangement. (b, c) image reconstructed with UBP and ADAS methods, respectively. (d, e) reconstructed image with applying ACF to UBP and ADAS, respectively. (f) 1D plot of PA amplitude of dashed-lines indicated in (a). All 2D images are normalized to their maximum and for better visibility we zero the negative parts of the reconstructed image in (a-e).

D. 2.25 MHz, 70% bandwidth detector with 20 dB SNR level scenario.

In the finite bandwidth and low SNR level scenario, reconstruction with UBP has a lower contrast ratio than the ADAS. This is because of band-limited detectors, DC and low-frequency components are degraded in ADAS and thus CR can be improved with this issue and the advantage of using the UBP method, which was to reduce the effect of low-frequency components, is degraded (see Fig. 4(a, c) and Fig. 4(b, d) in dB mode). Applying ACF to UBP and ADAS will improve the contrast ratio (see Fig. 4(e, g) and also, compare (b, d) with (f, h)). For better comparison and evaluation of the effect of applying ACF, a 1-D diagram of indicated dashed line in Fig. 4(a, c, e, g) is demonstrated in Fig. 4(i). Based on the shown 1D-line, the ADAS algorithm degrades background artifacts while negative artifacts have remained in UBP. Adding the ACF factor to the ADAS and UBP algorithm will reduce artifacts (both positive artifacts of the ADAS method and negative artifacts of the UBP method) and also lead to an increase in contrast ratio. The quantitative values of indicated dashed line are given in Table 3. Moreover, The FWHM of the imaging system's PSF without applying ACF to ADAS/UBP is 0.144 mm for both ADAS and UBP algorithms, and applying ACF to ADAS/UBP is about 0.12 mm for both of them for a point source located near the center. It can be concluded from this subsection that applying ACF improves the FWHM of the imaging system's PSF by more than 16% and reduces the resultant artifacts in the reconstructed image. The SNR value in reconstruction with ACF is 13.68 dB better than reconstruction without ACF for UBP while this increase is about 14.72 dB for ADAS+ACF.

TABLE 3: Quantitative values (CR) for 2.25 MHz, 70% bandwidth with 20 dB SNR level scenario.

Beamformer	UBP	ADAS	UBP+ACF	ADAS+ACF
CR (dB)	14.32	20.82	28	35.54

IV. DISCUSSION AND CONCLUSION

In this paper, we applied a fast coherence factor (CF) to the circular geometry photoacoustic tomography. Unlike the other methods, due to its simplicity, real-time and video-rate photoacoustic imaging can be achieved by applying this factor to the reconstructed image. As can be concluded from the performed simulations and obtained results, the use of ACF with UBP and ADAS algorithms is not dependent on the bandwidth and SNR level of the transducers and can be used for any desired detector with an SNR level.

ACF improves the tangential point-spread-function (PSF) and contrast ratio (CR) of the reconstructed image and reduces artifacts. Based on the quantitative values reported in tables (see Table 1, 2, 3), for different scenarios, ACF improves full-width-at half-maximum (FWHM) of the imaging system's PSF by 16%-20% and the CR by 13.68-30.24 dB, which are indicated in Table 4.

TABLE 4: CR and FWHM improvement values by applying ACF to different beamformer algorithms

Beamformer	UBP+ACF to UBP	ADAS+ACF to ADAS
CR improvement (dB) infinite BW and 80 dB SNR	30.24	15.46
CR improvement (%) infinite BW and 80 dB SNR	3150	494
FWHM improvement (%) infinite BW and 80 dB SNR	20	20
CR improvement (dB) infinite BW and 20 dB SNR	20.5	15.63
CR improvement (%) infinite BW and 20 dB SNR	959	604
FWHM improvement (%) infinite BW and 20 dB SNR	16	16
CR improvement (dB) finite BW and 20 dB SNR	13.68	14.72
CR improvement (%) finite BW and 20 dB SNR	383	548
FWHM improvement (%) finite BW and 20 dB SNR	16	16

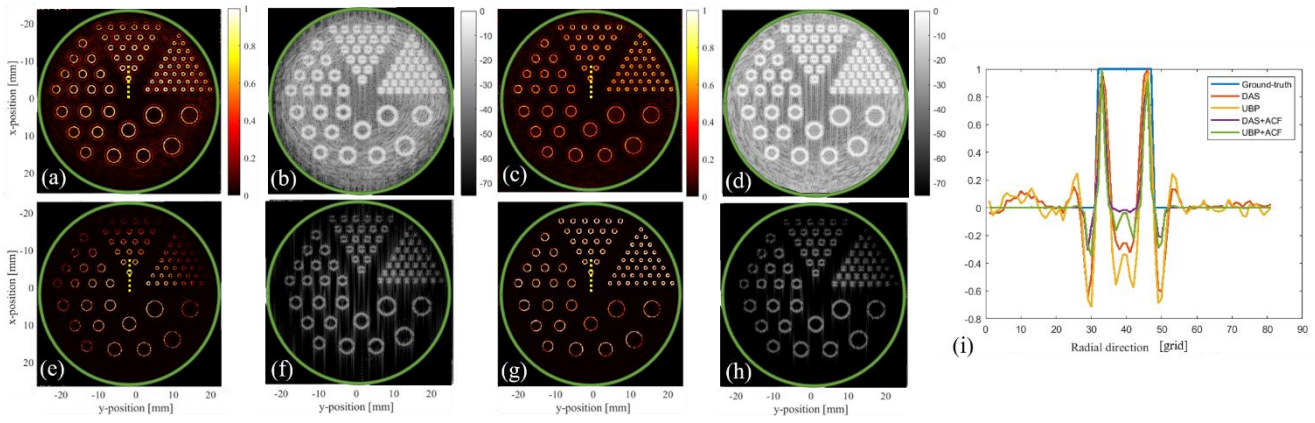


Fig. 4: Simulation results of 2.25 MHz, 70% bandwidth detector with 20 dB SNR level scenario. (a) initial Derenzo phantom and 360 point-like detectors with full-view circular arrangement. (b, c) image reconstructed with UBP and ADAS methods, respectively. (d, e) reconstructed image with applying ACF to UBP and ADAS, respectively. (f) 1D plot of PA amplitude of dashed-lines indicated in (a). all 2D images are normalized to their maximum and for better visibility we zero the negative parts of the reconstructed image in (a-e).

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